

Appendix: Uncertainty and Sentiment in the Filings of ARK ETF Holdings

Methods, supplementary figures and supplementary tables for the report of the same title. Section A gives every specification; Sections B and C hold the exhibits the report cites.

A. Methods

Word lists and parsing

Word lists are the entries with a positive year in the Negative and Uncertainty columns of the Loughran-McDonald Master Dictionary, 1993 to 2025 release, updated March 2026: 2,345 and 297 words. Filing text is the primary document with hidden inline-XBRL content removed and tables dropped where more than 15% of non-space characters are digits. Tokens are alphabetic strings of at least two characters, uppercased. Exhibits and material incorporated by reference are not recovered.

Scores

$$P_{cj} = \frac{\sum_{i \in c} tf_{ij}}{W_j}$$

$$T_{cj} = \sum_{i \in c, tf_{ij} > 0} \frac{1 + \ln(tf_{ij})}{1 + \ln(\bar{a}_j)} \ln\left(\frac{N}{df_i}\right)$$

P is the word share of list c in filing j and W the filing's word count. T is equation (1) of Loughran and McDonald (2011): tf is the count of word i in filing j, a the filing's average word frequency (words divided by distinct words), N the number of filings in the estimation corpus and df the number containing word i. Logarithms are natural. Document frequencies are fitted within each sample (ARK holdings, QQQ holdings), so tf.idf scores are comparable across filings within a sample and not across samples.

Locating Item 1A

The section opens at an occurrence of the heading Item 1A followed by Risk Factors that satisfies four conditions, and the last such occurrence in the filing is taken:

- It is not preceded within 60 characters by a reference cue such as in, see, under or Part I.
- It is not followed within 90 characters by another item number, which marks a table of contents or a cross-reference index.
- It is followed by text opening with a capital letter, since a citation continues with a lowercase word or a punctuation mark.
- It carries the item label; filers who head the section Risk Factors alone are recorded as not located.

The section closes at the next item heading whose title is also present, such as Item 1B Unresolved Staff Comments or Item 2 Unregistered Sales. A bare SIGNATURES line is not treated as a closing marker. Disclosure mode is read from the section's first 800 characters and its length: sections over 400 words are full restatements unless they claim no material change, in which case they are partial updates; shorter sections that claim no material change or point to the annual report are reference-only; shorter sections that do neither are recorded as nothing disclosed. Every statistic on the section is computed on the text sample, so the section exhibits and the tone exhibits describe the same filings.

Decomposition

$$S = w S_{\text{risk}} + (1 - w) S_{\text{body}}$$

$$\Delta S = \bar{w} \Delta S_{\text{risk}} + (1 - \bar{w}) \Delta S_{\text{body}} + \Delta w (\bar{S}_{\text{risk}} - \bar{S}_{\text{body}})$$

w is Item 1A's share of the filing's words and bars denote the mean over the two filings being compared. The first two terms are the language term and the third the composition term; the identity is exact. Comparisons are within company and within form. The language term inherits the high uncertainty density of a one-sentence pointer, so the change in the filing excluding Item 1A is the statistic to read (Table C11).

Event windows and controls

Day 0 is the first NYSE session on or after the later of the filing date and the EDGAR acceptance date, with acceptance at or after the close moved to the next session. The filing-period return is the adjusted close on day 3 over the adjusted close on day -1, less the same ratio for SPY. Prior volatility is the sample standard deviation of 55 daily returns over days -60 to -6, annualised; post-filing volatility uses 60 returns over days 4 to 63. Size is the nominal close on day -1 times the cover-page share count of the filing being scored, from the XBRL fact EntityCommonStockSharesOutstanding. Dollar volume is the mean of nominal close times volume over days -60 to -6. Prior excess return is the SPY-adjusted buy-and-hold return over the same window. Filings with a day -1 price below three dollars are excluded. The outcome regressions carry company and calendar-quarter effects, log size, log dollar volume, prior excess return and, where stated, prior volatility.

Regressions and inference

The aggregate trend model regresses the quarterly mean on elapsed years and three seasonal indicators, with ordinary and Newey-West (four lags) standard errors. The within-company model adds company effects to filing-level observations. Saturated calendar-quarter effects are not used in trend models because they absorb the trend. Standard errors in all filing-level regressions are clustered two ways by company and calendar quarter, with inference degrees of freedom one fewer than the smaller cluster count; where that covariance is not positive definite the report marks the cell and uses company clusters alone. Effects per standard deviation use the standard deviation of the measure in the estimation sample. The detectable effect is the coefficient that a two-sided 5% test would find with 80% power at the estimated standard error. An indicator regressor identified by fewer than ten filings is reported as not estimable.

The comparison with QQQ holdings

QQQ tracks the Nasdaq-100; its holdings are the 102 constituents of 9 September 2026, which the same filters reduce to 94 SEC filers. Each result in Section 7 of the report is a single regression on the companies held by only one portfolio, 22 companies held by both being excluded so that no filing sits on both sides: the coefficient on a QQQ indicator for a level difference (calendar-quarter effects, seasonal indicators), on its interaction with elapsed years for a trend difference, and on its interaction with the measure for a difference in an outcome coefficient (company and calendar-quarter effects, the outcome regressions' controls). A significant estimate in one portfolio beside an insignificant one in the other is never read as a difference. QQQ's own exhibits, computed on all its holdings including the shared companies, are Tables C4 to C7.

B. Supplementary figures

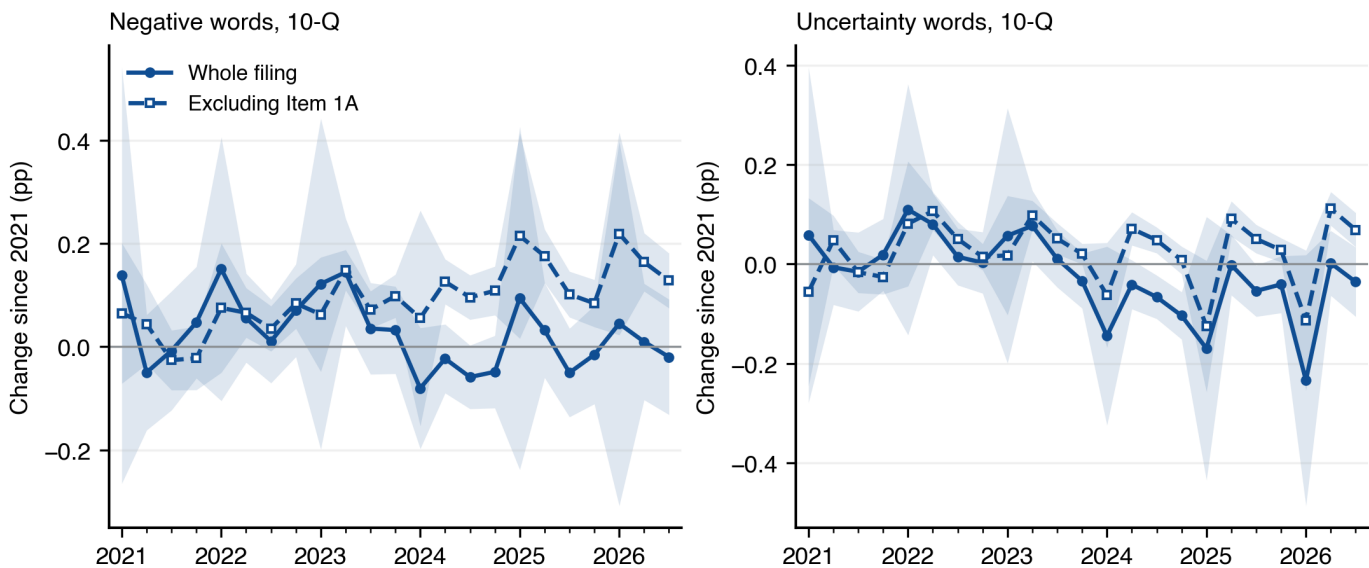


Figure B1. ARK quarterly reports, whole filing (solid) and excluding Item 1A (dashed), company-centred means as change since 2021 with 95% bands. The whole-filing line is flat while the line excluding Item 1A rises for negative words (Table 8).

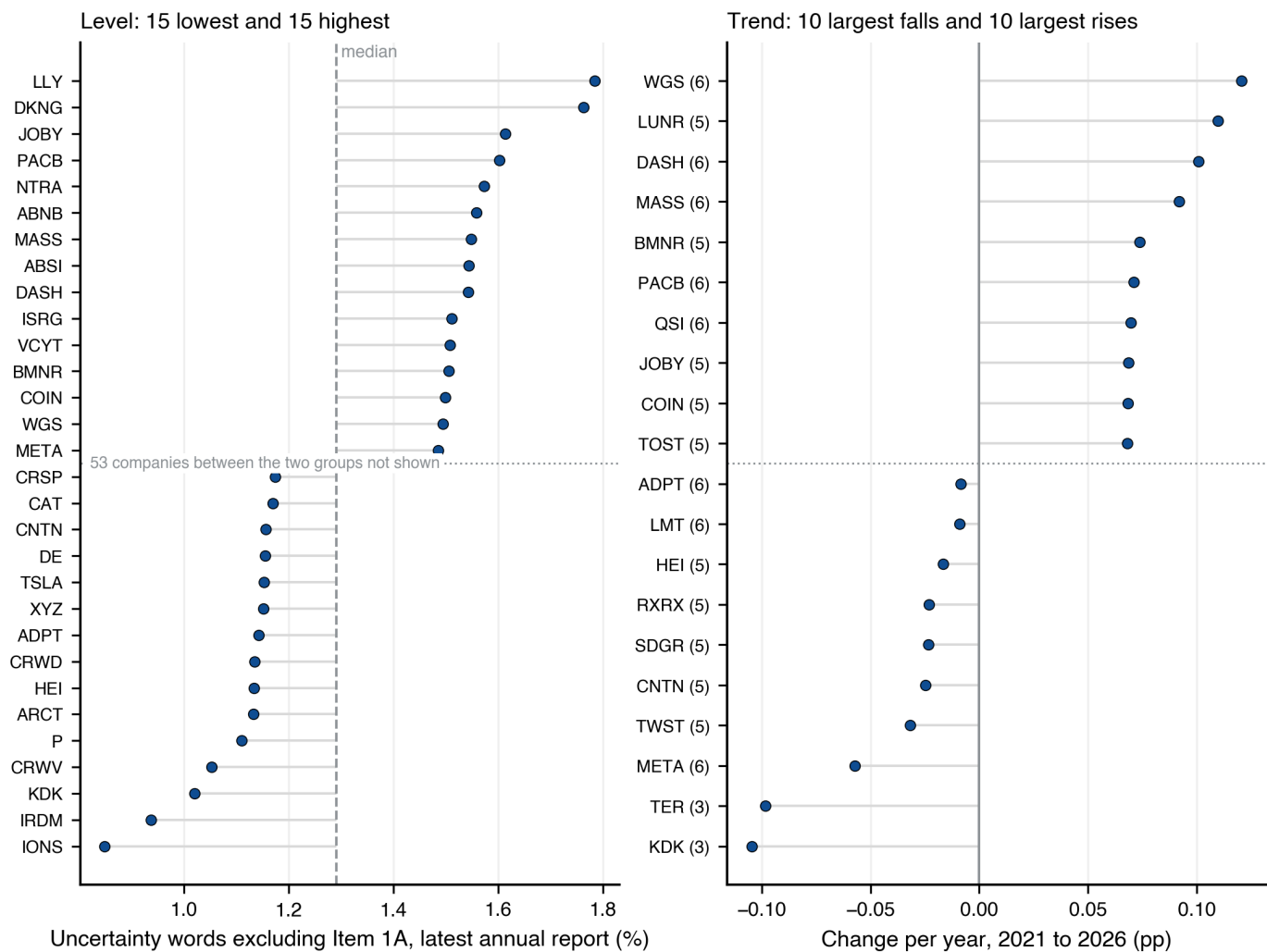


Figure B2. ARK holdings ranked on uncertainty words excluding Item 1A. Left: latest annual report with a located Item 1A, 15 lowest and 15 highest of 83 companies; dashed line, the median. Right: within-company slope over 2021 to 2026, the 10 largest falls and 10 largest rises among the 71 companies with at least three such reports (number of reports in brackets); slopes rest on three to six observations and rank companies. Table C8 lists every company.

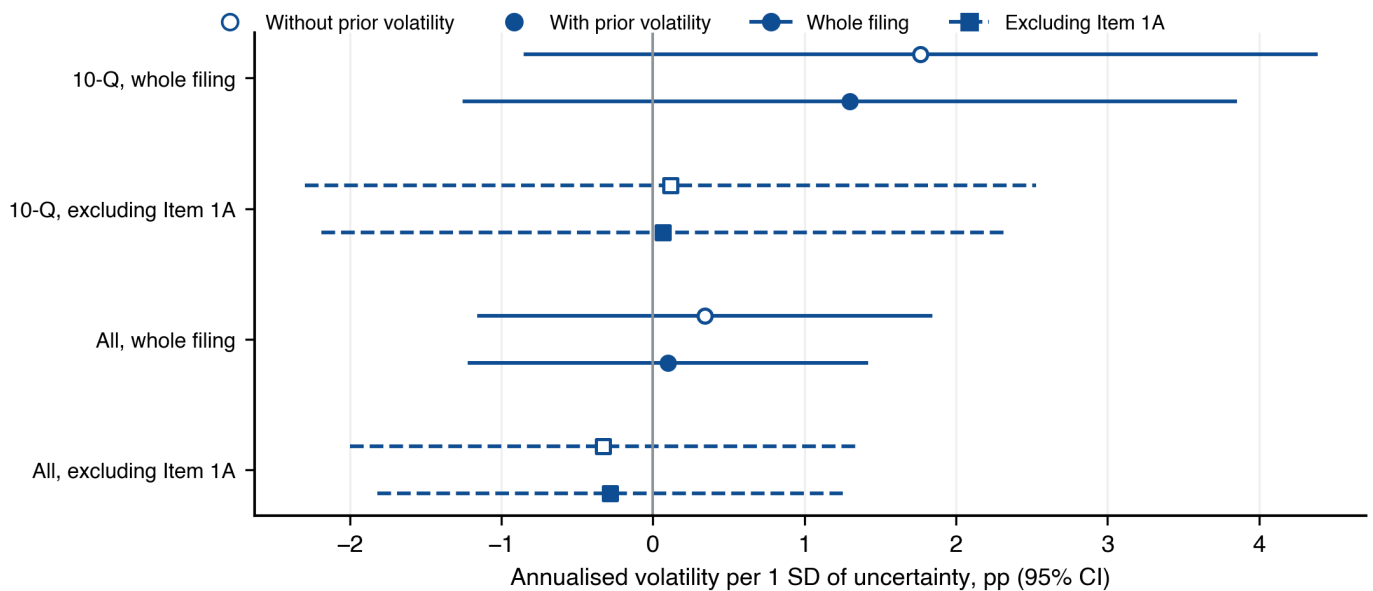


Figure B3. Effect of one standard deviation of uncertainty word share on annualised volatility with 95% intervals, ARK quarterly reports and all reports with a located Item 1A, without (hollow) and with (filled) prior volatility, whole filing (circles) and excluding Item 1A (squares). Annual reports alone are omitted; their intervals span tens of percentage points.

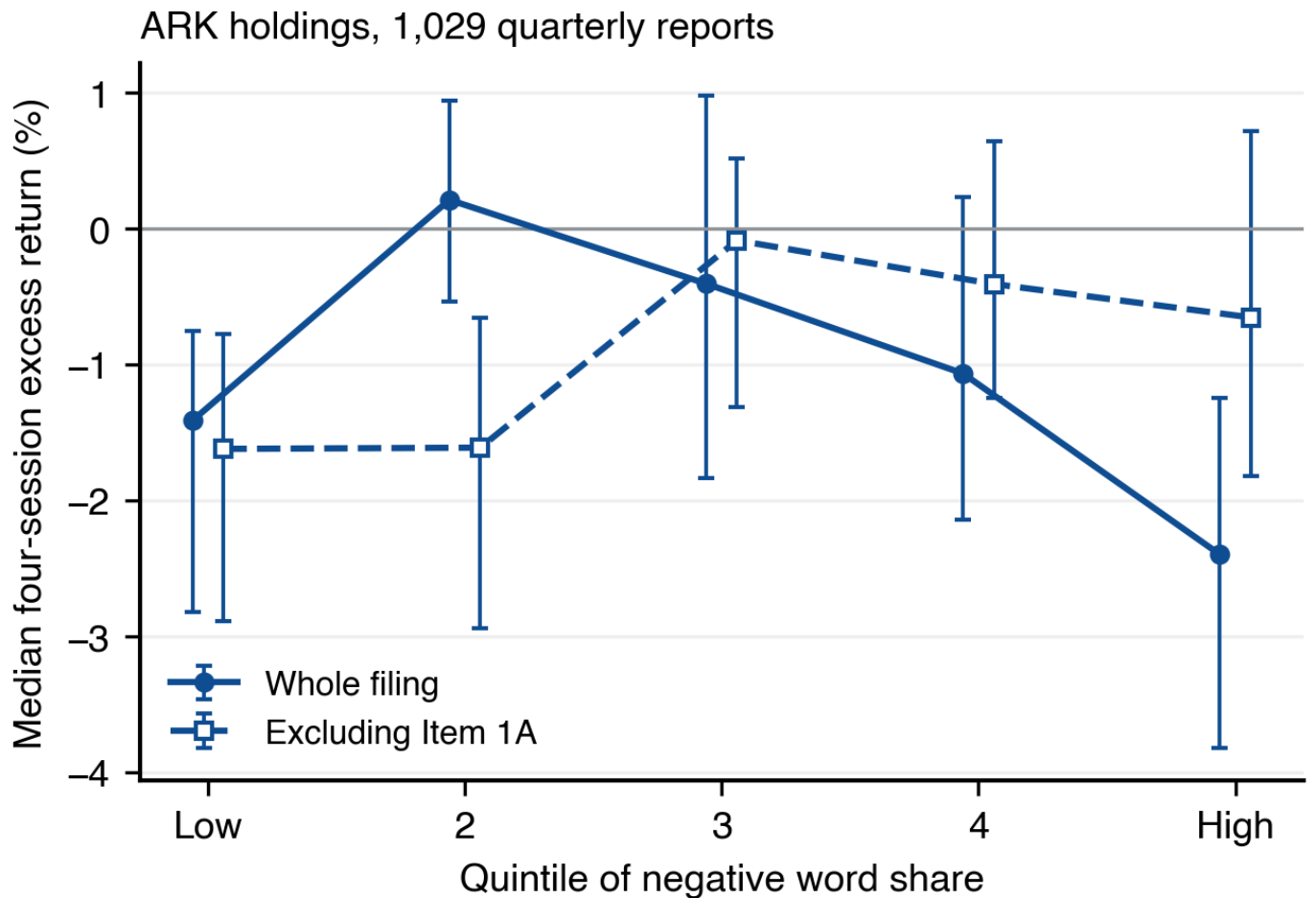


Figure B4. Median four-session excess return by quintile of negative word share, ARK quarterly reports with a located Item 1A and complete market data, whole filing and excluding Item 1A, with bootstrap 95% intervals. The design follows Figure 1 of Loughran and McDonald (2011).

- Risk factors restated
- No material change, updates given
- No material change, reader referred to the annual report
- Nothing disclosed

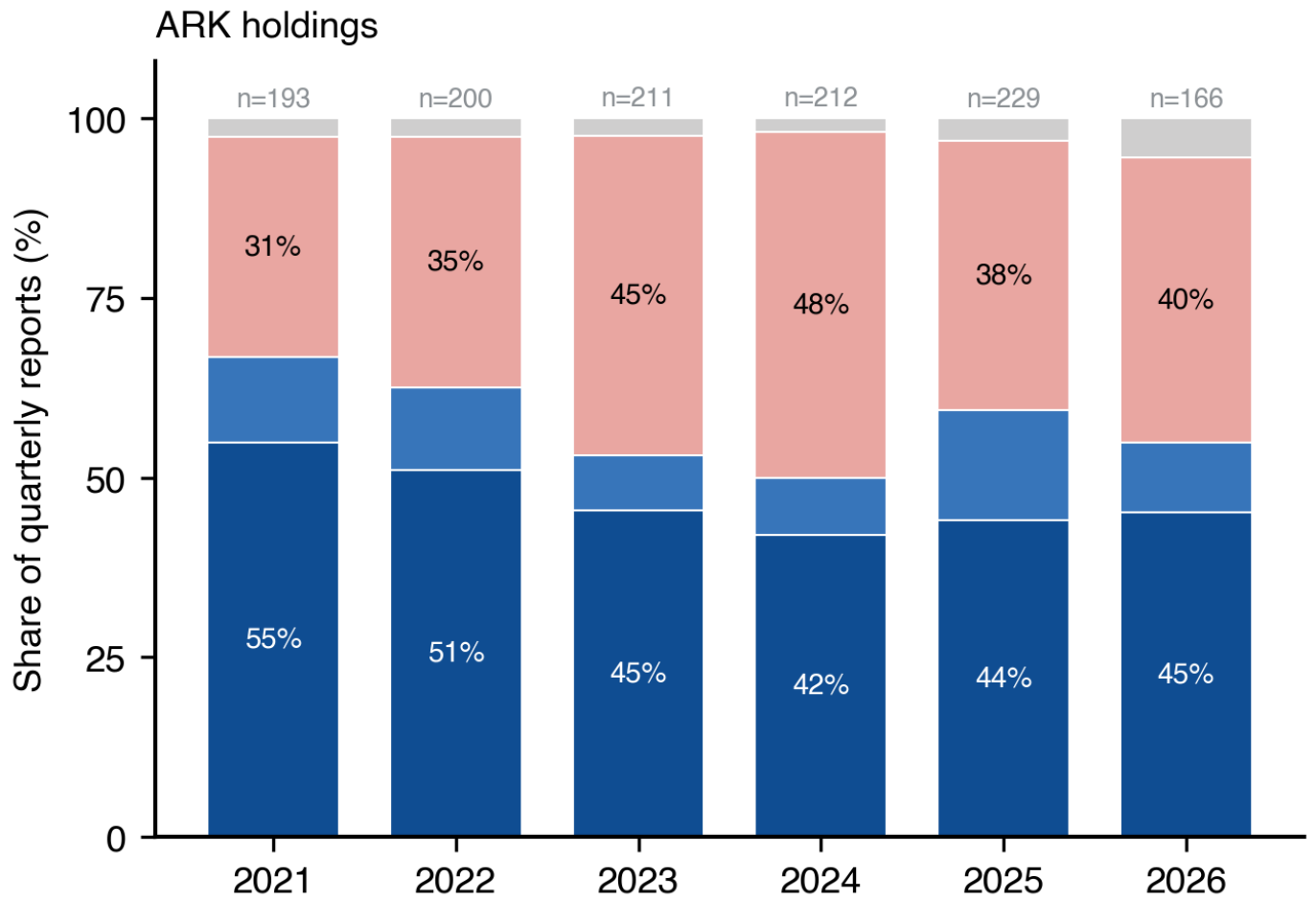


Figure B5. Disclosure mode of Item 1A in ARK quarterly reports with a located heading, by filing year; 2026 covers filings through September.

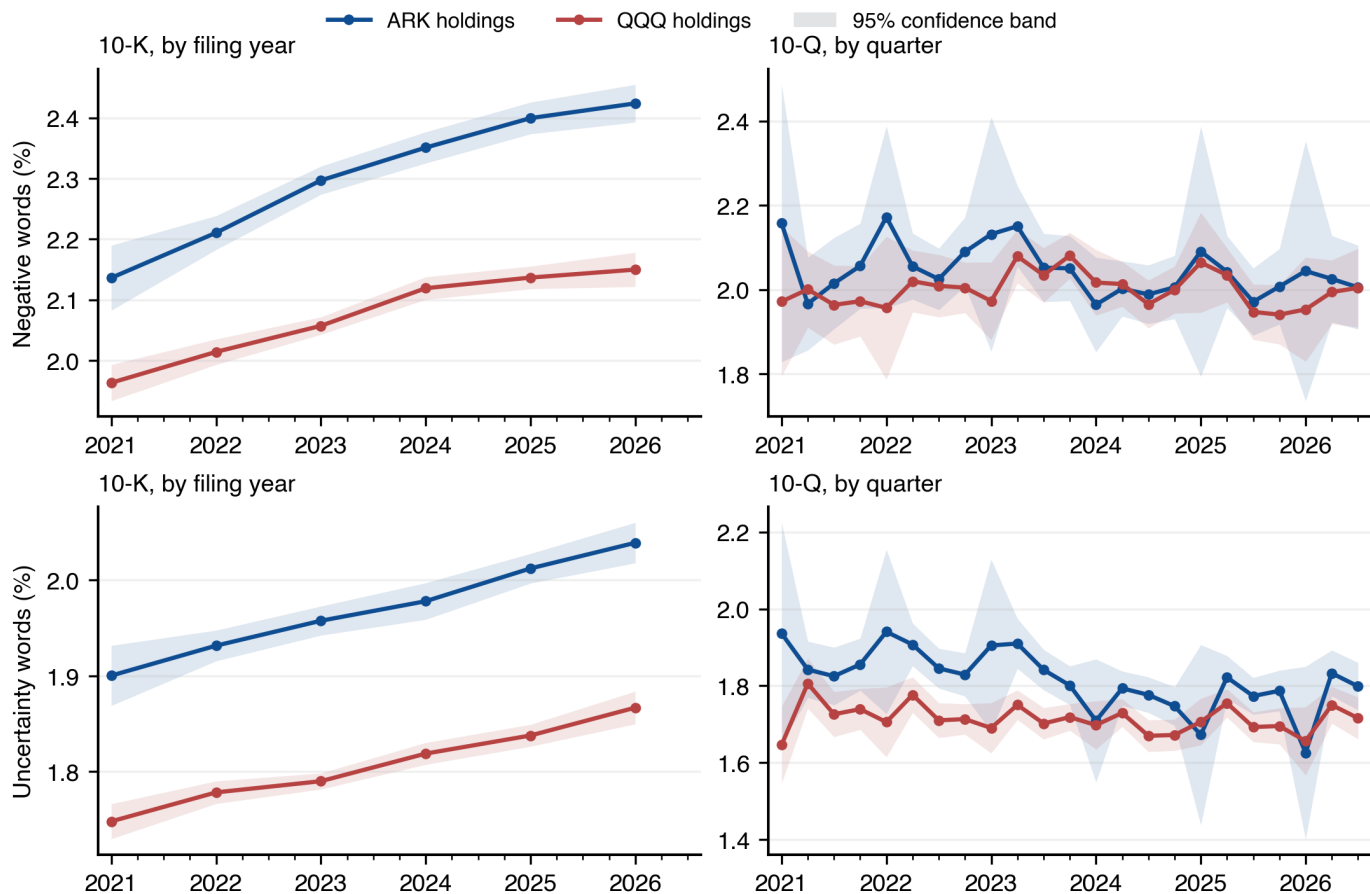


Figure B6. Both measures by quarter, ARK holdings and QQQ holdings (each including the shared companies), company-centred means with 95% bands; annual reports by filing year.

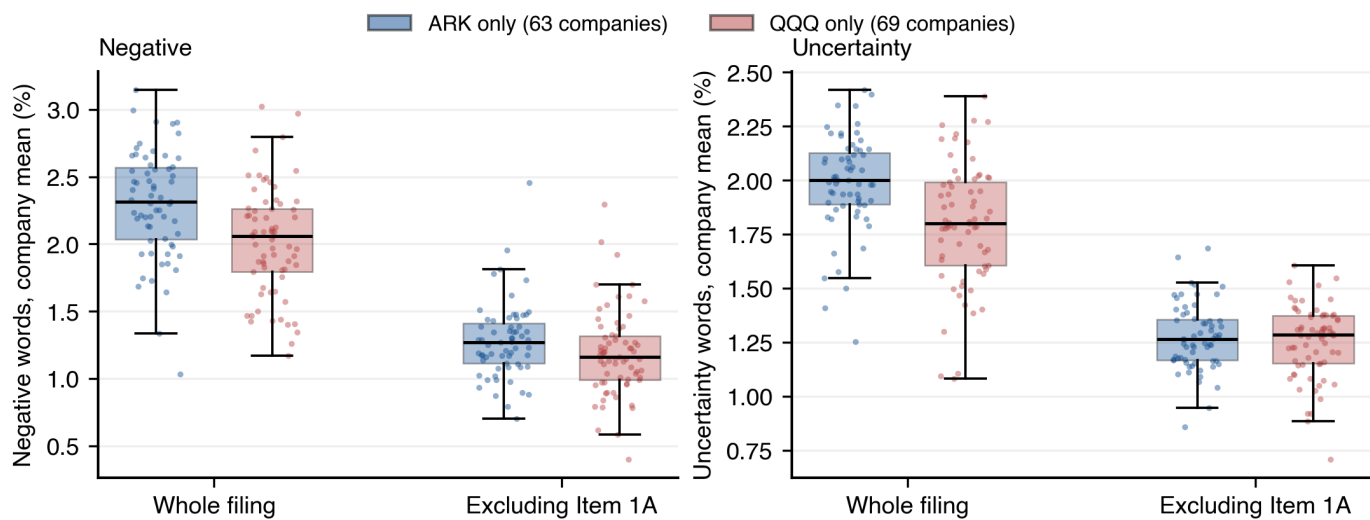


Figure B7. Company-mean word shares in annual reports with a located Item 1A, ARK-only against QQQ-only companies, whole filing and excluding Item 1A. Boxes span the interquartile range, whiskers the 5th to 95th percentile, dots are companies. The level rows of Table 9 test these differences.

C. Supplementary tables

Table C1. Disclosure mode by year, quarterly reports (%)

Shares of all quarterly reports in the text sample, including those with no located heading.

Portfolio	Year	Reports	Risk factors restated	No material change, updates given	No material change, reader referred to the annual report	Nothing disclosed	No Item 1A heading located
ARK holdings	2021	209	50.7	11.0	28.2	2.4	7.7
ARK holdings	2022	224	45.5	10.3	31.2	2.2	10.7
ARK holdings	2023	233	41.2	6.9	40.3	2.1	9.4
ARK holdings	2024	237	37.6	7.2	43.0	1.7	10.5
ARK holdings	2025	253	39.9	13.8	34.0	2.8	9.5
ARK holdings	2026	187	40.1	8.6	35.3	4.8	11.2
QQQ holdings	2021	255	50.6	6.3	34.5	5.1	3.5
QQQ holdings	2022	257	49.0	10.9	26.5	8.6	5.1
QQQ holdings	2023	260	41.5	8.1	34.6	8.5	7.3
QQQ holdings	2024	261	39.1	4.6	41.0	7.3	8.0
QQQ holdings	2025	271	41.7	7.7	36.5	6.3	7.7
QQQ holdings	2026	190	39.5	10.0	36.8	6.8	6.8

Table C2. ARK holdings: disclosure switches and market outcomes

Same controls and clustering as Tables 5 and 6. Treated: filings with the indicator on. An indicator identified by fewer than ten filings is not estimated.

Report	Outcome	Regressor	Filings	Treated	Estimate (pp)	SE	p	Status
All	Four-session excess return	Stopped restating	1,257	69	+2.86	1.75	0.118	
All	Post-filing volatility, prior controlled	Stopped restating	1,189	66	-0.03	2.94	0.991	
All	Four-session excess return	Started restating	1,257	60	-0.37	1.65	0.825	
All	Post-filing volatility, prior controlled	Started restating	1,189	55	-1.32	1.89	0.493	
All	Four-session excess return	Restates this quarter	1,257	415	-0.23	0.93	0.806	
All	Post-filing volatility, prior controlled	Restates this quarter	1,189	385	-0.17	2.05	0.933	
10-K	Four-session excess return	Stopped restating	282				n.e.	not estimable
10-K	Post-filing volatility, prior controlled	Stopped restating	281				n.e.	not estimable
10-K	Four-session excess return	Started restating	282				n.e.	not estimable
10-K	Post-filing volatility, prior controlled	Started restating	281				n.e.	not estimable
10-K	Four-session excess return	Restates this quarter	282				n.e.	not estimable
10-K	Post-filing volatility, prior controlled	Restates this quarter	281				n.e.	not estimable
10-Q	Four-session excess return	Stopped restating	975	69	+3.00	1.68	0.089	
10-Q	Post-filing volatility, prior controlled	Stopped restating	908	66	-1.49	3.30	0.655	

Report	Outcome	Regressor	Filings	Treated	Estimate (pp)	SE	p	Status
10-Q	Four-session excess return	Started restating	975	60	-0.06	1.82	0.973	
10-Q	Post-filing volatility, prior controlled	Started restating	908	55	-2.09	2.00	0.308	
10-Q	Four-session excess return	Restates this quarter	975	415	-1.00	1.04	0.347	
10-Q	Post-filing volatility, prior controlled	Restates this quarter	908	385	+1.02	2.70	0.709	

Table C3. ARK holdings: aggregate trend, ordinary against Newey-West standard errors

Report	Measure	Standard errors	Slope per year	t	p
10-K	Negative, word share	Ordinary	+0.061	8.94	< 0.001
10-K	Negative, word share	Newey-West, 4 lags	+0.061	9.68	< 0.001
10-K	Uncertainty, word share	Ordinary	+0.024	4.02	0.001
10-K	Uncertainty, word share	Newey-West, 4 lags	+0.024	4.48	< 0.001
10-K	Negative, tf.idf	Ordinary	+2.992	2.43	0.030
10-K	Negative, tf.idf	Newey-West, 4 lags	+2.992	2.80	0.015
10-K	Uncertainty, tf.idf	Ordinary	-0.231	-1.05	0.313
10-K	Uncertainty, tf.idf	Newey-West, 4 lags	-0.231	-1.23	0.241
10-Q	Negative, word share	Ordinary	-0.015	-1.42	0.173
10-Q	Negative, word share	Newey-West, 4 lags	-0.015	-1.77	0.093
10-Q	Uncertainty, word share	Ordinary	-0.033	-3.02	0.007
10-Q	Uncertainty, word share	Newey-West, 4 lags	-0.033	-3.71	0.002
10-Q	Negative, tf.idf	Ordinary	-1.243	-1.57	0.134
10-Q	Negative, tf.idf	Newey-West, 4 lags	-1.243	-1.57	0.134
10-Q	Uncertainty, tf.idf	Ordinary	-0.431	-3.24	0.005
10-Q	Uncertainty, tf.idf	Newey-West, 4 lags	-0.431	-3.13	0.006

Table C4. QQQ holdings: summary statistics by report type

tf.idf fitted on the QQQ corpus.

Form	Measure	N	Mean	SD	P25	Median	P75
10-K	Negative, word share	518	2.073	0.438	1.790	2.065	2.373
10-K	Uncertainty, word share	518	1.806	0.292	1.623	1.824	2.019
10-K	Negative, tf.idf	518	139.315	49.956	104.510	132.289	164.273
10-K	Uncertainty, tf.idf	518	24.671	7.381	19.308	24.377	29.231
10-Q	Negative, word share	1,494	2.003	0.884	1.208	1.825	2.759
10-Q	Uncertainty, word share	1,494	1.722	0.594	1.239	1.540	2.238
10-Q	Negative, tf.idf	1,494	72.226	56.303	23.310	51.265	114.335
10-Q	Uncertainty, tf.idf	1,494	12.715	6.845	6.938	11.322	17.435

Table C5. QQQ holdings: trend tests, aggregate and within company

Report	Measure	Aggregate slope	Newey-West t	Within-company slope	t	p	Filings
10-K	Negative, word share	+0.052	9.95	+0.041	6.64	< 0.001	518
10-K	Uncertainty, word share	+0.027	4.02	+0.024	8.45	< 0.001	518
10-K	Negative, tf.idf	+2.263	4.13	+1.294	3.14	0.005	518
10-K	Uncertainty, tf.idf	-0.262	-2.29	-0.200	-2.04	0.053	518
10-Q	Negative, word share	-0.012	-1.89	-0.002	-0.13	0.900	1,494
10-Q	Uncertainty, word share	-0.013	-6.49	-0.008	-0.85	0.404	1,494
10-Q	Negative, tf.idf	-1.134	-8.70	-0.750	-0.74	0.468	1,494
10-Q	Uncertainty, tf.idf	-0.510	-5.64	-0.463	-3.39	0.003	1,494

Table C6. QQQ holdings: post-filing volatility on uncertainty

Report	Measure	Prior volatility	Filings	Effect per SD (pp)	Detectable (pp)	p
All	Uncertainty, word share	No	1,863	-0.09	1.64	0.871
All	Uncertainty, word share	Yes	1,863	-0.07	1.49	0.899
All	Uncertainty, tf.idf	No	1,863	-1.36	2.97	0.194
All	Uncertainty, tf.idf	Yes	1,863	-1.40	2.82	0.162
10-K	Uncertainty, word share	No	489	+0.30	6.25	0.889
10-K	Uncertainty, word share	Yes	489	-0.08	4.66	0.960
10-K	Uncertainty, tf.idf	No	489	-3.17	6.56	0.173
10-K	Uncertainty, tf.idf	Yes	489	-2.98	5.63	0.137
10-Q	Uncertainty, word share	No	1,374	+0.24	2.52	0.781
10-Q	Uncertainty, word share	Yes	1,374	+0.17	2.27	0.832
10-Q	Uncertainty, tf.idf	No	1,374	+0.33	3.31	0.776
10-Q	Uncertainty, tf.idf	Yes	1,374	+0.25	3.02	0.809

Table C7. QQQ holdings: four-session excess return on negative tone

Report	Measure	Filings	Effect per SD (pp)	Detectable (pp)	p
All	Negative, word share	1,947	+0.26	0.74	0.323
All	Negative, tf.idf	1,947	+0.78	1.12	0.055
10-K	Negative, word share	499	+1.20	4.92	0.486
10-K	Negative, tf.idf	499	+1.34	5.22	0.463
10-Q	Negative, word share	1,448	+0.46	1.57	0.401
10-Q	Negative, tf.idf	1,448	+0.94	2.28	0.241

Table C8. ARK holdings: uncertainty words excluding Item 1A, every company

Filed: the latest annual report with a located Item 1A. Slopes over all such reports, where at least three exist.

Ticker	Filed	Latest annual report (%)	Slope per year (pp)	Annual reports
LLY	2026-02-12	1.78	+0.041	6
DKNG	2026-02-13	1.76	+0.035	4
JOBY	2026-02-27	1.61	+0.069	5
PACB	2026-02-25	1.60	+0.071	6
NTRA	2026-02-27	1.57	+0.036	6

Ticker	Filed	Latest annual report (%)	Slope per year (pp)	Annual reports
ABNB	2026-02-12	1.56	+0.038	6
MASS	2026-03-09	1.55	+0.092	6
ABSI	2026-03-24	1.54	+0.044	5
DASH	2026-02-18	1.54	+0.101	6
ISRG	2026-02-03	1.51	+0.003	6
VCYT	2026-02-26	1.51	+0.004	6
BMNR	2025-11-21	1.51	+0.074	5
COIN	2026-02-12	1.50	+0.068	5
WGS	2026-02-23	1.49	+0.121	6
META	2026-01-29	1.49	-0.057	6
TDY	2026-02-20	1.48	+0.011	6
CERS	2021-02-25	1.47		
KTOS	2026-02-23	1.47	+0.007	6
GH	2026-02-19	1.45	+0.010	6
LMT	2026-01-29	1.45	-0.009	6
TXG	2026-02-13	1.44	+0.063	6
NFLX	2026-01-23	1.44	+0.027	6
SHOP	2026-02-11	1.43		
TOST	2026-02-18	1.43	+0.068	5
Z	2026-02-11	1.43	+0.063	6
CRCL	2026-03-09	1.42		
DDOG	2026-02-18	1.41	+0.020	5
AVGO	2025-12-18	1.39	-0.008	5
QSI	2026-03-03	1.38	+0.070	6
RKLB	2026-02-26	1.38	+0.059	3
GOOG	2026-02-05	1.36	+0.006	6
PSNL	2022-02-24	1.36		
HOOD	2026-02-18	1.36	+0.038	5
NET	2026-02-26	1.35	+0.025	6
LUNR	2026-03-19	1.35	+0.110	5
BWXT	2026-02-23	1.34	-0.005	6
RXRX	2026-02-25	1.34	-0.023	5
CDNA	2026-02-25	1.34	+0.011	6
OKLO	2026-03-17	1.33	+0.041	5
SNPS	2025-12-22	1.32	+0.024	5
BEAM	2026-02-24	1.30	+0.020	3
GTLB	2026-03-17	1.29	-0.000	5
AVAV	2026-06-29	1.29	-0.003	6
AMZN	2026-02-06	1.29	-0.004	6
RBRK	2026-03-19	1.27		
TWST	2025-11-17	1.26	-0.032	5
NTLA	2026-02-26	1.26		
SDGR	2026-02-25	1.25	-0.023	5
GRMN	2021-02-17	1.25		
RBLX	2026-02-11	1.24	+0.001	5
NVDA	2026-02-25	1.24	+0.005	6
CMPS	2026-03-24	1.24	+0.014	5
SYM	2025-11-24	1.24	+0.034	5
AUR	2026-02-11	1.23	-0.006	5
MELI	2026-02-25	1.23	-0.003	6
TER	2023-02-22	1.23	-0.098	3
SNOW	2026-03-20	1.22	+0.052	6
TRMB	2026-02-25	1.21	+0.006	6

Ticker	Filed	Latest annual report (%)	Slope per year (pp)	Annual reports
ACHR	2026-03-02	1.21	+0.018	6
FRNM	2026-03-12	1.21		
BFLY	2026-02-27	1.20	+0.028	6
AMD	2026-02-04	1.19	+0.039	6
SOFI	2026-02-17	1.19	+0.012	5
LHX	2026-02-12	1.19	+0.004	6
PRME	2026-03-03	1.19	+0.009	4
FIG	2026-02-18	1.18		
PLTR	2026-02-17	1.18	+0.056	6
NRIX	2026-01-28	1.18	+0.009	5
CRSP	2021-02-16	1.17		
CAT	2026-02-13	1.17	+0.028	6
CNTN	2026-03-31	1.16	-0.025	5
DE	2025-12-18	1.16	+0.025	5
TSLA	2026-01-29	1.15	+0.036	6
XYZ	2026-02-26	1.15	+0.012	6
ADPT	2026-02-26	1.14	-0.008	6
CRWD	2026-03-05	1.14	+0.027	6
HEI	2025-12-22	1.13	-0.017	5
ARCT	2022-03-01	1.13		
P	2026-03-25	1.11	+0.001	6
CRWV	2026-03-02	1.05		
KDK	2026-03-11	1.02	-0.105	3
IRDM	2026-02-12	0.94	+0.012	6
IONS	2026-02-26	0.85	+0.002	6

Table C9. ARK holdings: the 20 largest disclosure switches in quarterly reports, of 158

Ranked by the change in Item 1A words. The full list is in the data workbook.

Ticker	Filed	Transition	Item 1A words before	Item 1A words after	Uncertainty change (pp)	Change excl. Item 1A (pp)
PRME	2023-08-07	Stopped restating	49,292	75	-1.573	-0.285
COIN	2026-05-07	Stopped restating	42,072	138	-1.004	+0.051
TXG	2025-08-08	Stopped restating	39,630	57	-1.625	-0.323
TOST	2024-05-08	Stopped restating	33,026	188	-1.225	+0.229
RXR	2022-05-10	Stopped restating	32,188	26	-1.232	-0.070
ABSI	2022-05-11	Stopped restating	31,789	134	-1.096	+0.128
VCYT	2025-05-08	Stopped restating	31,137	57	-0.815	-0.095
TXG	2022-05-05	Stopped restating	29,175	55	-0.882	+0.098
QSI	2021-11-15	Stopped restating	26,336	64	-1.263	-0.049
MASS	2022-05-10	Stopped restating	26,115	74	-1.045	+0.164
AUR	2024-05-09	Stopped restating	23,941	67	-1.221	+0.316
TSLA	2023-04-24	Stopped restating	13,216	43	-0.882	-0.012
ACHR	2021-08-16	Started restating	28	12,007	+0.868	+0.041
AUR	2022-05-12	Started restating	138	21,823	+1.815	+0.500
MASS	2021-11-04	Started restating	74	26,115	+1.201	-0.010
QSI	2021-08-16	Started restating	130	26,336	+1.300	+0.132
WGS	2021-11-15	Started restating	131	34,121	+1.415	+0.159
SOFI	2021-08-16	Started restating	134	36,200	+0.941	+0.301
ABSI	2025-05-13	Started restating	134	38,588	+1.587	+0.389
TXG	2025-05-09	Started restating	52	39,630	+1.433	+0.128

Table C10. Share of filings with a located Item 1A heading

Group	10-K (%)	10-Q (%)
ARK only	87.8	87.1
Both	93.6	100.0
QQQ only	92.9	91.8

Table C11. Filing-to-filing change in word share by disclosure transition

Quarterly reports, each compared with the same company's previous quarterly report. The language and composition terms sum to the change.

Portfolio	Measure	Transition	Filings	Companies	Change (pp)	Change excl. Item 1A (pp)	Language term (pp)	Composition term (pp)	Median change in Item 1A words
ARK holdings	Negative	Kept restating	567	67	+0.009	+0.010	+0.013	-0.004	157
ARK holdings	Negative	Kept referring	398	48	-0.000	-0.000	+0.000	-0.000	0
ARK holdings	Negative	Started restating	77	33	+0.344	+0.039	+0.146	+0.198	668
ARK holdings	Negative	Stopped restating	81	36	-0.380	-0.003	-0.167	-0.213	-835
ARK holdings	Uncertainty	Kept restating	567	67	+0.001	+0.004	+0.004	-0.003	157
ARK holdings	Uncertainty	Kept referring	398	48	-0.009	-0.008	-0.009	-0.000	0
ARK holdings	Uncertainty	Started restating	77	33	+0.230	+0.025	-0.201	+0.431	668
ARK holdings	Uncertainty	Stopped restating	81	36	-0.207	+0.052	+0.346	-0.553	-835
QQQ holdings	Negative	Kept restating	659	64	+0.009	-0.002	+0.004	+0.005	101
QQQ holdings	Negative	Kept referring	528	52	-0.016	-0.016	-0.016	+0.000	0
QQQ holdings	Negative	Started restating	61	33	+0.282	+0.006	+0.156	+0.126	551
QQQ holdings	Negative	Stopped restating	59	34	-0.265	+0.011	-0.108	-0.158	-581
QQQ holdings	Uncertainty	Kept restating	659	64	-0.000	-0.002	-0.004	+0.004	101
QQQ holdings	Uncertainty	Kept referring	528	52	-0.007	-0.007	-0.007	+0.000	0
QQQ holdings	Uncertainty	Started restating	61	33	+0.142	-0.032	-0.169	+0.311	551
QQQ holdings	Uncertainty	Stopped restating	59	34	-0.160	+0.052	+0.218	-0.378	-581

Table C12. QQQ less ARK: levels and trends, whole filing and excluding Item 1A

One regression per row on the disjoint sample, filings with a located Item 1A. Two-way clustered inference; Table 9 falls back to company clusters where the status column is marked.

Report	Measure	Scope	Quantity	QQQ less ARK	p	Filings	Status
All	Negative	whole filing	Level (pp)	-0.1178	0.335	2,653	
All	Negative	whole filing	Trend (pp a year)	-0.0231	0.234	2,653	
All	Negative	excluding Item 1A	Level (pp)	+0.0244	0.753	2,653	
All	Negative	excluding Item 1A	Trend (pp a year)	-0.0392	0.014	2,653	
All	Uncertainty	whole filing	Level (pp)	-0.1407	0.074	2,653	
All	Uncertainty	whole filing	Trend (pp a year)	-0.0136	0.233	2,653	

Report	Measure	Scope	Quantity	QQQ less ARK	p	Filings	Status
All	Uncertainty	excluding Item 1A	Level (pp)	-0.0614	0.088	2,653	
All	Uncertainty	excluding Item 1A	Trend (pp a year)	-0.0154	0.060	2,653	
All	Section length	Item 1A share of words	Level (pp)	-5.7572	0.114	2,653	
All	Section length	Item 1A share of words	Trend (pp a year)	+0.1423	0.794	2,653	
10-K	Negative	whole filing	Level (pp)	-0.2733	0.003	682	
10-K	Negative	whole filing	Trend (pp a year)	-0.0155	0.222	682	
10-K	Negative	excluding Item 1A	Level (pp)	-0.0512	0.383	682	
10-K	Negative	excluding Item 1A	Trend (pp a year)	-0.0340	n.e.	682	covariance not positive definite
10-K	Uncertainty	whole filing	Level (pp)	-0.2005	< 0.001	682	
10-K	Uncertainty	whole filing	Trend (pp a year)	-0.0070	0.001	682	
10-K	Uncertainty	excluding Item 1A	Level (pp)	-0.0080	0.786	682	
10-K	Uncertainty	excluding Item 1A	Trend (pp a year)	-0.0068	n.e.	682	covariance not positive definite
10-K	Section length	Item 1A share of words	Level (pp)	-7.9891	0.006	682	
10-K	Section length	Item 1A share of words	Trend (pp a year)	+0.0880	0.814	682	
10-Q	Negative	whole filing	Level (pp)	-0.0811	0.583	1,971	
10-Q	Negative	whole filing	Trend (pp a year)	-0.0268	0.310	1,971	
10-Q	Negative	excluding Item 1A	Level (pp)	+0.0620	0.485	1,971	
10-Q	Negative	excluding Item 1A	Trend (pp a year)	-0.0413	0.008	1,971	
10-Q	Uncertainty	whole filing	Level (pp)	-0.1401	0.149	1,971	
10-Q	Uncertainty	whole filing	Trend (pp a year)	-0.0146	0.414	1,971	
10-Q	Uncertainty	excluding Item 1A	Level (pp)	-0.0777	0.060	1,971	
10-Q	Uncertainty	excluding Item 1A	Trend (pp a year)	-0.0141	0.148	1,971	
10-Q	Section length	Item 1A share of words	Level (pp)	-5.8511	0.193	1,971	
10-Q	Section length	Item 1A share of words	Trend (pp a year)	+0.1612	0.843	1,971	

Table C13. QQQ less ARK: difference in the volatility and return coefficients

Interaction of the measure with a QQQ indicator, one regression per row on the disjoint sample; all filings for the rows so marked, otherwise filings with a located Item 1A.

Outcome	Report	Scope	Measure	Prior volatility	QQQ less ARK (pp per SD)	p	Filings
Post-filing volatility	All	all filings	Uncertainty, word share	No	-0.77	0.427	2,616
Post-filing volatility	All	all filings	Uncertainty, word share	Yes	-0.32	0.694	2,616
Post-filing volatility	All	all filings	Uncertainty, tf.idf	No	-1.43	0.314	2,616
Post-filing volatility	All	all filings	Uncertainty, tf.idf	Yes	-1.27	0.312	2,616
Post-filing volatility	All	whole filing	Uncertainty, word share	No	-0.75	0.447	2,352
Post-filing volatility	All	whole filing	Uncertainty, word share	Yes	-0.31	0.715	2,352

Outcome	Report	Scope	Measure	Prior volatility	QQQ less ARK (pp per SD)	p	Filings
Post-filing volatility	All	excluding Item 1A	Uncertainty, word share	No	-0.38	0.780	2,352
Post-filing volatility	All	excluding Item 1A	Uncertainty, word share	Yes	-0.26	0.830	2,352
Post-filing volatility	10-K	all filings	Uncertainty, word share	No	+11.64	0.004	689
Post-filing volatility	10-K	all filings	Uncertainty, word share	Yes	+10.54	0.007	689
Post-filing volatility	10-K	all filings	Uncertainty, tf.idf	No	-5.78	0.358	689
Post-filing volatility	10-K	all filings	Uncertainty, tf.idf	Yes	-4.80	0.384	689
Post-filing volatility	10-K	whole filing	Uncertainty, word share	No	+11.06	< 0.001	627
Post-filing volatility	10-K	whole filing	Uncertainty, word share	Yes	+10.04	0.002	627
Post-filing volatility	10-K	excluding Item 1A	Uncertainty, word share	No	+0.63	0.891	627
Post-filing volatility	10-K	excluding Item 1A	Uncertainty, word share	Yes	-0.72	0.863	627
Post-filing volatility	10-Q	all filings	Uncertainty, word share	No	-3.35	0.046	1,927
Post-filing volatility	10-Q	all filings	Uncertainty, word share	Yes	-2.52	0.105	1,927
Post-filing volatility	10-Q	all filings	Uncertainty, tf.idf	No	-4.27	0.065	1,927
Post-filing volatility	10-Q	all filings	Uncertainty, tf.idf	Yes	-3.43	0.111	1,927
Post-filing volatility	10-Q	whole filing	Uncertainty, word share	No	-2.83	0.128	1,725
Post-filing volatility	10-Q	whole filing	Uncertainty, word share	Yes	-2.10	0.237	1,725
Post-filing volatility	10-Q	excluding Item 1A	Uncertainty, word share	No	-1.32	0.491	1,725
Post-filing volatility	10-Q	excluding Item 1A	Uncertainty, word share	Yes	-1.11	0.522	1,725
Four-session excess return	All	all filings	Negative, word share		+0.95	0.197	2,742
Four-session excess return	All	all filings	Negative, tf.idf		+0.66	0.391	2,742
Four-session excess return	All	whole filing	Negative, word share		+1.24	0.116	2,466
Four-session excess return	All	excluding Item 1A	Negative, word share		-0.73	0.376	2,466
Four-session excess return	10-K	all filings	Negative, word share		+1.11	0.716	700
Four-session excess return	10-K	all filings	Negative, tf.idf		+2.62	0.583	700
Four-session excess return	10-K	whole filing	Negative, word share		+0.42	0.898	637
Four-session excess return	10-K	excluding Item 1A	Negative, word share		+0.32	0.896	637
Four-session excess return	10-Q	all filings	Negative, word share		+2.30	0.054	2,042
Four-session excess return	10-Q	all filings	Negative, tf.idf		+2.95	0.035	2,042
Four-session excess return	10-Q	whole filing	Negative, word share		+2.91	0.007	1,829
Four-session excess return	10-Q	excluding Item 1A	Negative, word share		-1.03	0.450	1,829

Loughran-McDonald Master Dictionary, 1993 to 2025 release, updated March 2026.

U.S. Securities and Exchange Commission, EDGAR filings and company facts.